

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- a. We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | \mathbf{x}_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | \mathbf{x}_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

$$H_0: \sigma_M^2 = \sigma_F^2 \quad H_A: \sigma_M^2 \neq \sigma_F^2, \quad \alpha = 0.05$$

$$N_{\text{male}} = 577 \Rightarrow N_{\text{female}} = 1000 - 577 = 423$$

$$\hat{\sigma}_F = 12.024 \Rightarrow \hat{\sigma}_F^2 = 12.024^2 = \frac{SSE_F}{N - K} = \frac{SSE_F}{577 - 4}$$

$$\Rightarrow SSE_F = 60577.59$$

$$\hat{\sigma}_M^2 = \frac{SSE_M}{N - K} = \frac{97161.9174}{423 - 4} = 169.57$$

$$F^* = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.57}{144.58} \approx 1.173 \sim F(573, 419)$$

$$\alpha \text{ critical value } F(573, 419, \overset{\text{左尾面积}}{0.05}) \approx 0.8377$$

$$F(573, 419, \underline{0.975}) \approx 1.1968$$

$$\Rightarrow RR = \{F^* \mid F^* < 0.8377 \text{ or } F^* > 1.1968\}$$

Since $F^* = 1.173$ 不在 RR 之中 $\Rightarrow$ We do not reject $H_0: \sigma_M^2 = \sigma_F^2$

- b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | x_i, \text{MARRIED} = 0) = \sigma_{\text{SINGLE}}^2$ and $\text{var}(e_i | x_i, \text{MARRIED} = 1) = \sigma_{\text{MARRIED}}^2$. Specify the null hypothesis $\sigma_{\text{SINGLE}}^2 = \sigma_{\text{MARRIED}}^2$ versus the alternative hypothesis $\sigma_{\text{MARRIED}}^2 > \sigma_{\text{SINGLE}}^2$. We add *FEMALE* to the wage equation as an explanatory variable, so that

$$\text{WAGE}_i = \beta_1 + \beta_2 \text{EDUC}_i + \beta_3 \text{EXPER}_i + \beta_4 \text{METRO}_i + \beta_5 \text{FEMALE} + e_i \quad (\text{XR8.6b})$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

$$H_0: \sigma_{\text{single}}^2 = \sigma_{\text{married}}^2 \quad H_a: \sigma_{\text{single}}^2 \neq \sigma_{\text{married}}^2 \quad \alpha = 0.05 \quad \begin{matrix} N_{\text{single}} = 400 \\ N_{\text{married}} = 600 \end{matrix}$$

$$\hat{\sigma}_{\text{single}}^2 = \frac{\text{SSE}_{\text{single}}}{N - k} = \frac{56231.0382}{400 - 5} \approx 141.16$$

$$\hat{\sigma}_{\text{married}}^2 = \frac{\text{SSE}_{\text{married}}}{N - k} = \frac{100703.0471}{600 - 5} \approx 169.15$$

$$F^* = \frac{\hat{\sigma}_{\text{married}}^2}{\hat{\sigma}_{\text{single}}^2} = \frac{169.15}{141.16} \approx 1.189 \sim F(1595, 195)$$

$$\text{critical value } F(1595, 195, 0.025) = 0.8538$$

$$F(1595, 195, 0.975) = 1.1954$$

$$\Rightarrow \text{RR} = \{F^* \mid F^* < 0.8538 \text{ or } F^* > 1.1954\}$$

Since $F^* = 1.189$ is not in RR, we do not reject

$$H_0: \sigma_{\text{single}}^2 = \sigma_{\text{married}}^2$$

- c. Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.

$$NR^2 = 59.03 \sim \chi^2_{(5-1)}, \alpha = 0.05$$

$$\text{critical value } \chi^2_{(4, 0.05)} = 9.49$$

$$\Rightarrow RR = \{ \chi^2 \mid \chi^2 > 9.49 \}$$

Since $NR^2 = 59.03 > 9.49$, which is in the RR

$\Rightarrow$ We can reject H_0 , and 推論存在異質變異

- d. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

$$NR^2 = 78.82$$

$\therefore$ Explanatory variables 共 $4 + 4 + C_2^4 = 14$ (不含 intercept)

$$\Rightarrow NR^2 = 78.82 \sim \chi^2_{14}$$

$$\chi^2 \text{ critical value} = \chi^2_{(14, 0.05)} = 23.68$$

$$\Rightarrow RR = \{ \chi^2 \mid \chi^2 > 23.68 \}$$

Since $NR^2 = 78.82 > 23.68$, which is in RR

$\Rightarrow$ We can reject H_0 , 推論存在異質變異*

e. The OLS fitted model from part (b), with usual and robust standard errors, is

$$\widehat{WAGE} = -17.77 + 2.50EDUC + 0.23EXPER + 3.23METRO - 4.20FEMALE$$

(se)	(2.36)	(0.14)	(0.031)	(1.05)	(0.81)
(robse)	(2.50)	(0.16)	(0.029)	(0.84)	(0.80)

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

{
Narrow
Interval estimate :
wider

EXPER, METRO, FEMALE
EDUC

There is no inconsistency in the result.

∴ Model 中存在異質變異，使得有些 explanatory variables 的 se 會被錯誤地高估，有些會被錯誤地低估，但 robust se 會矯正這些錯誤，使原先高估的 se ↓ 以及原先低估的 se ↑

- f. If we add *MARRIED* to the model in part (b), we find that its *t*-value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

This is not conflict with the result in (b).

Since GQ Test 是檢測 model 中有無異質變異

t Test 是檢測該 variable 是否顯著

Therefore, *t*-value = 1 僅說明 *MARRIED* 對 *WAGE*

沒有顯著影響